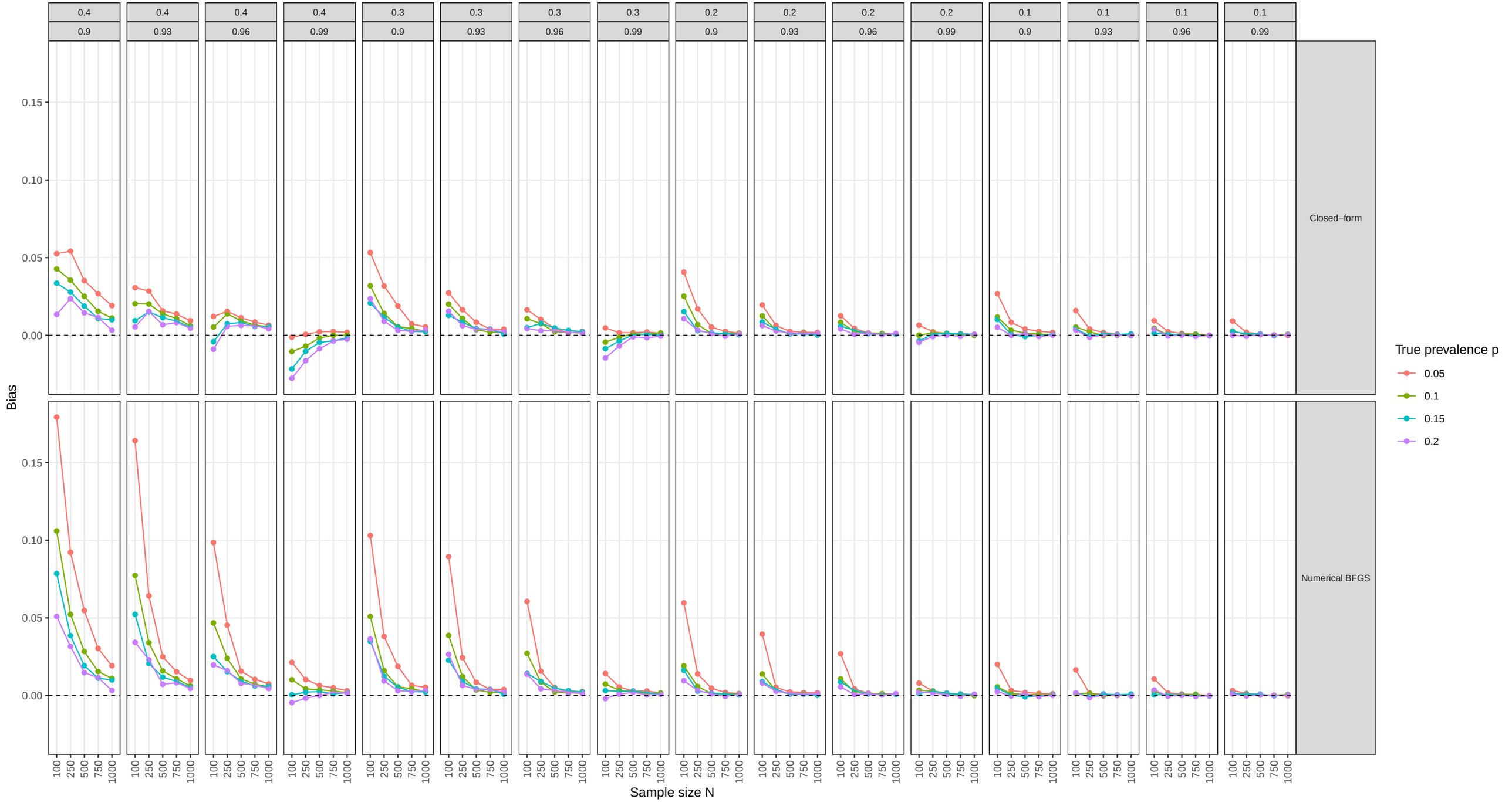
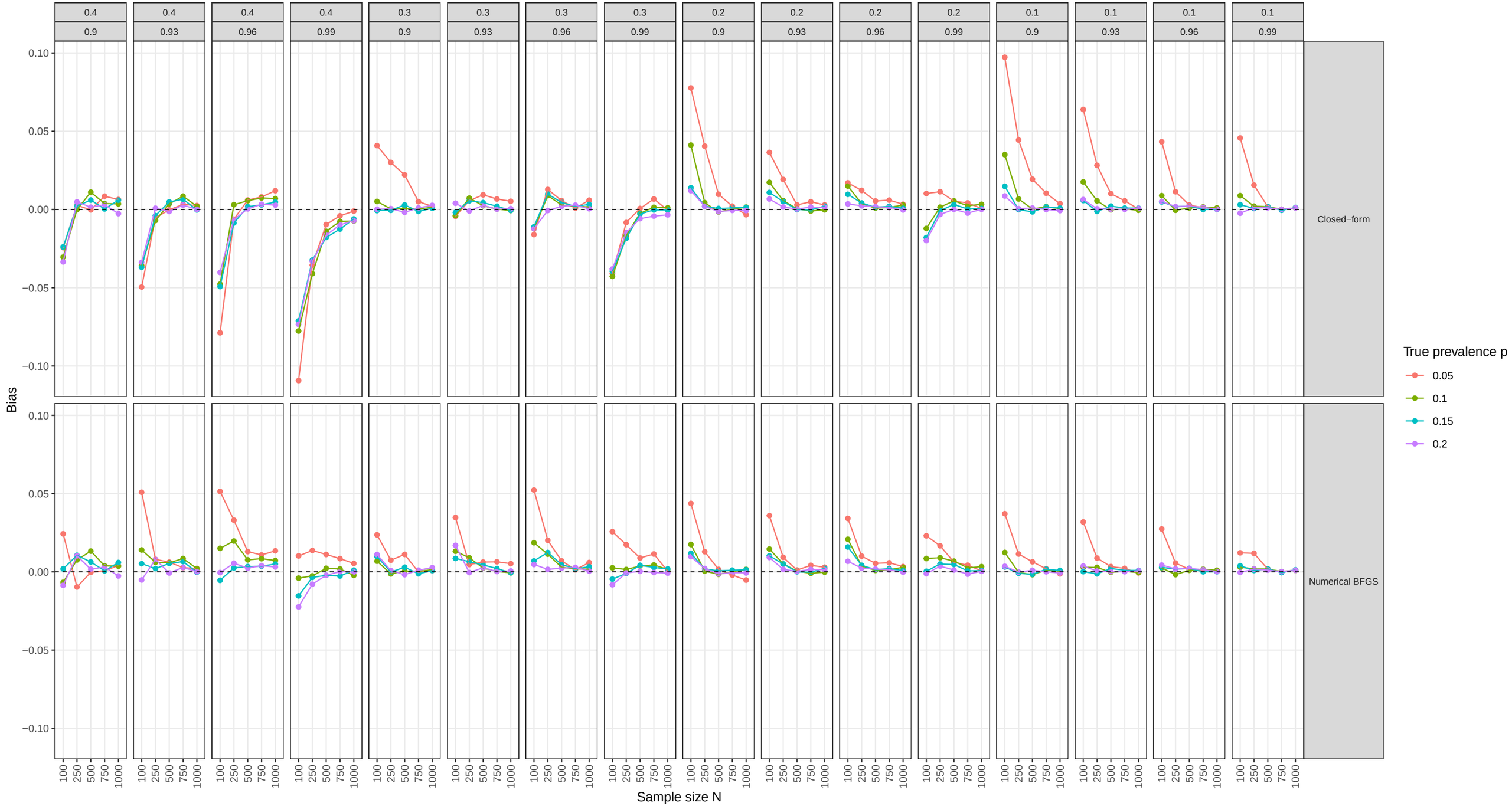
Bias for prevalence p

Columns show alpha and gamma; all prevalence values are displayed together



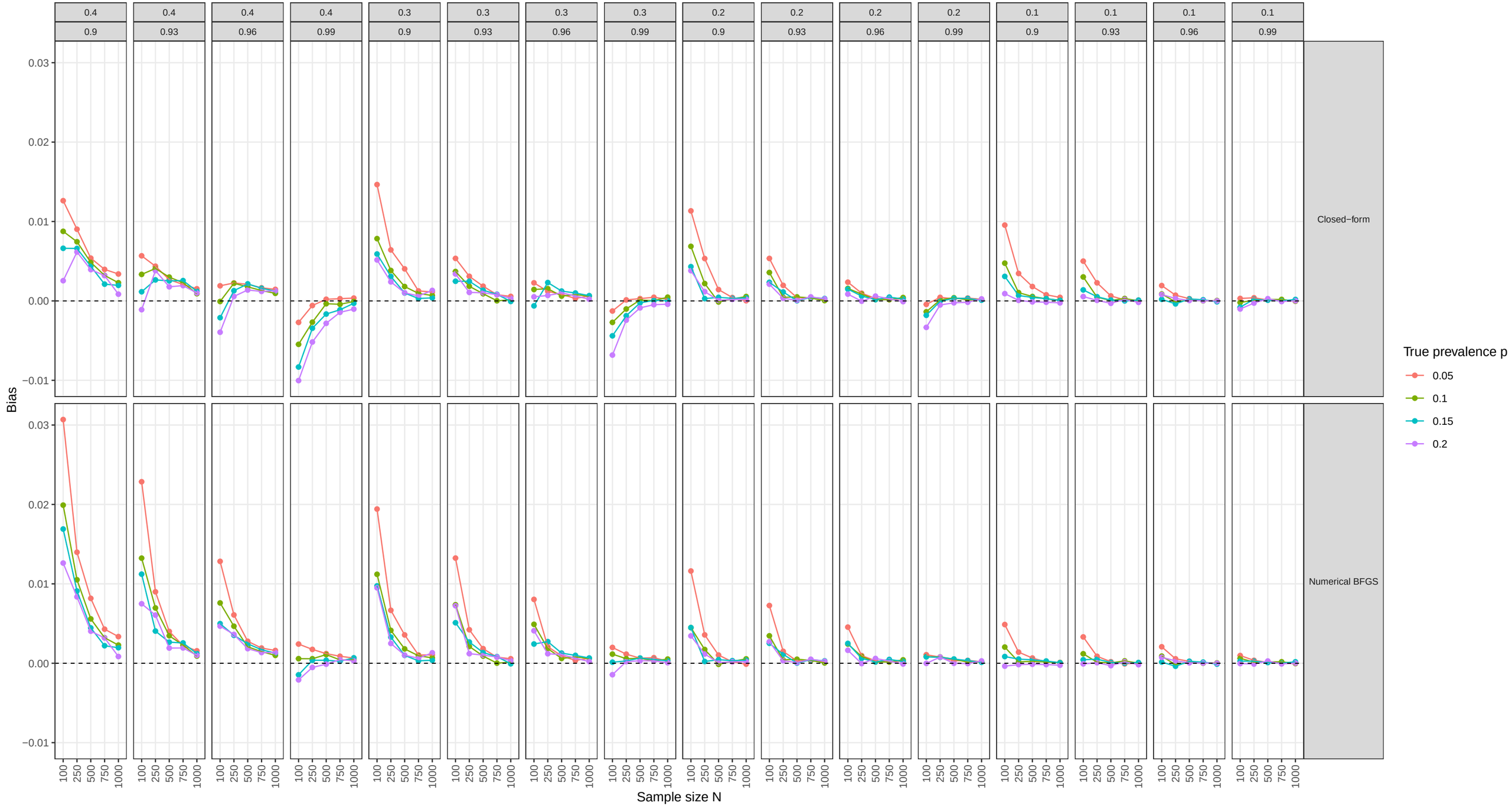
Bias for false−negative probability alpha

Columns show alpha and gamma; all prevalence values are displayed together



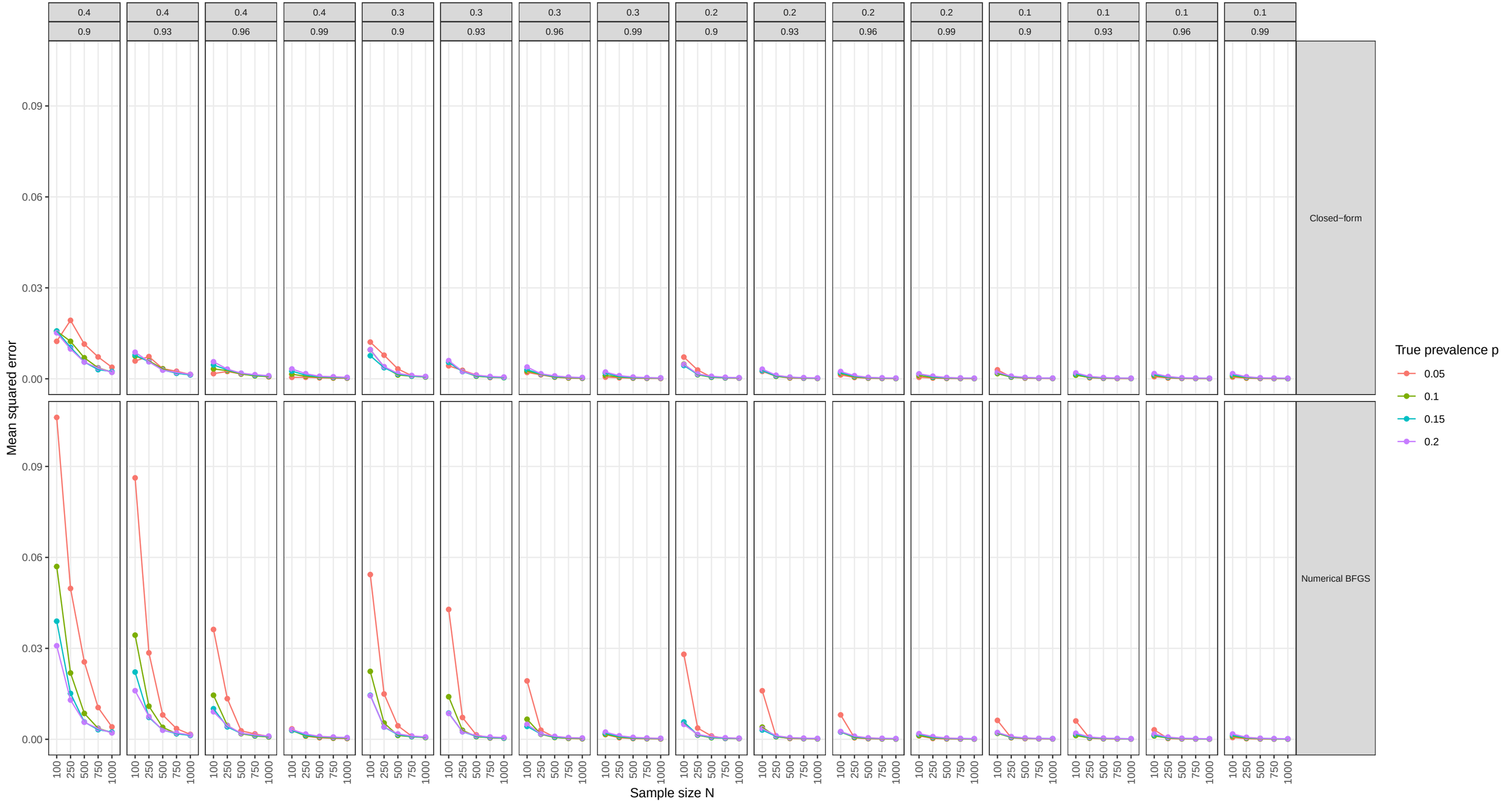
Bias for specificity gamma

Columns show alpha and gamma; all prevalence values are displayed together



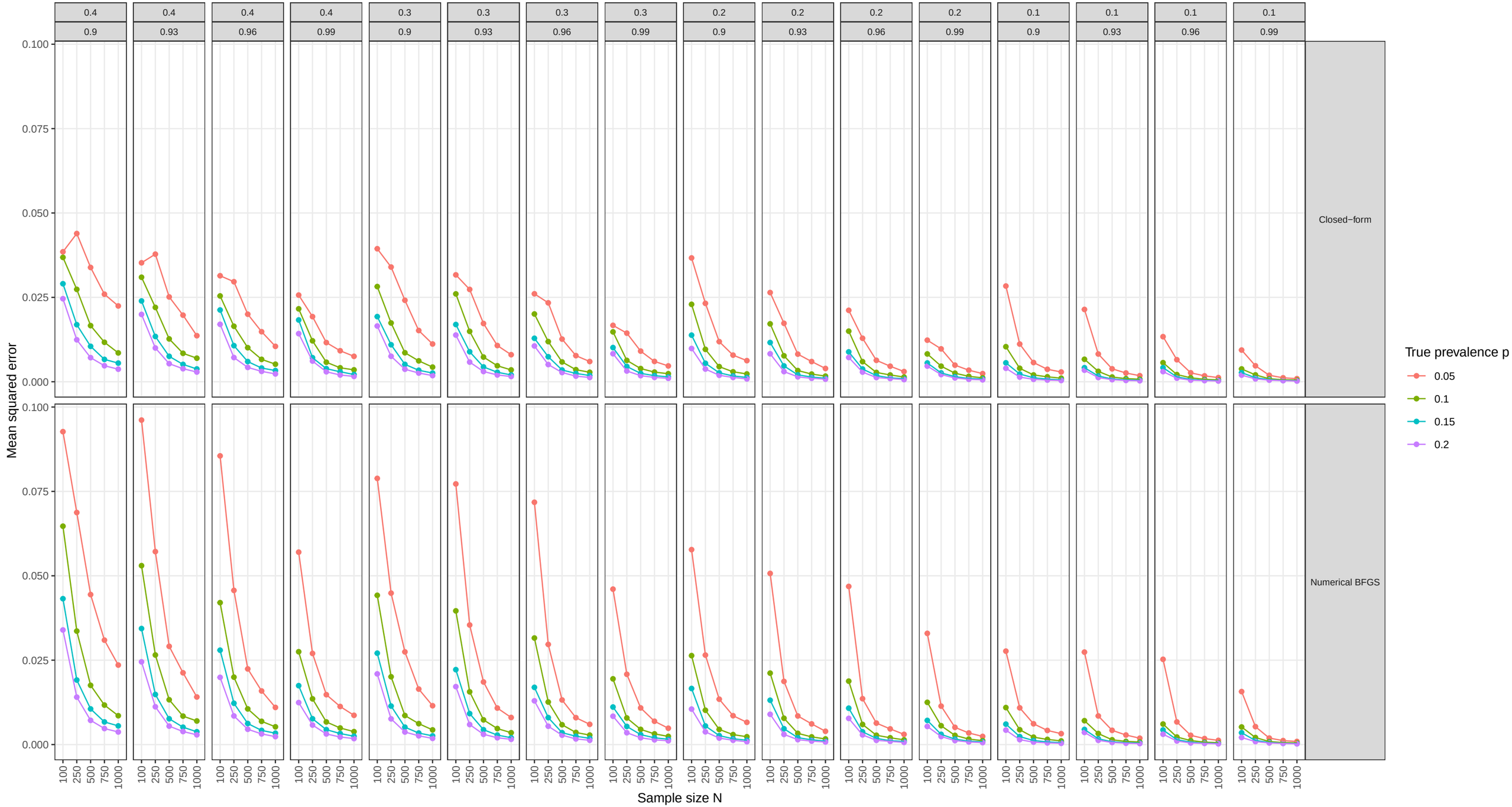
MSE for prevalence p

Columns show alpha and gamma; all prevalence values are displayed together



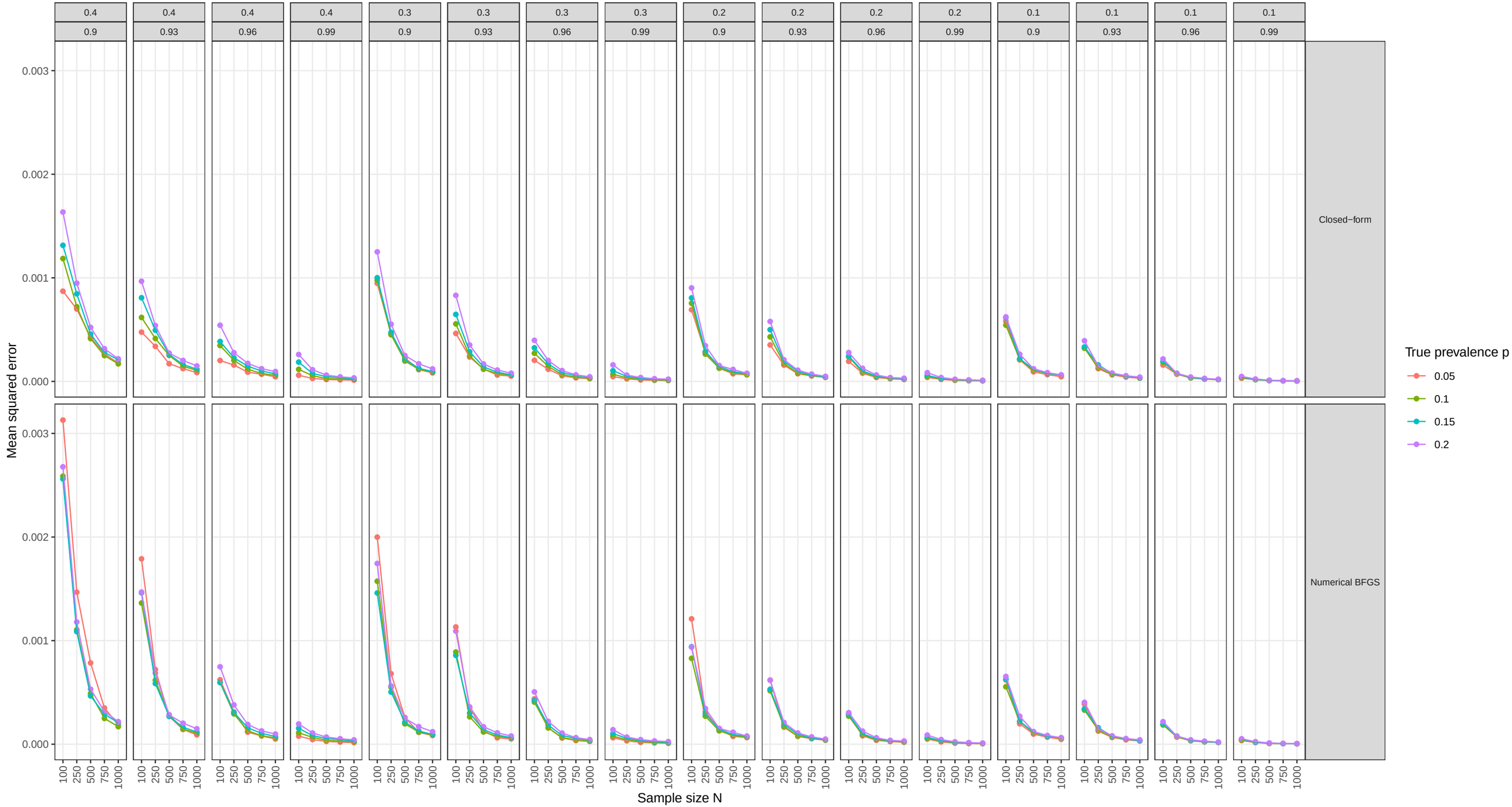
MSE for false−negative probability alpha

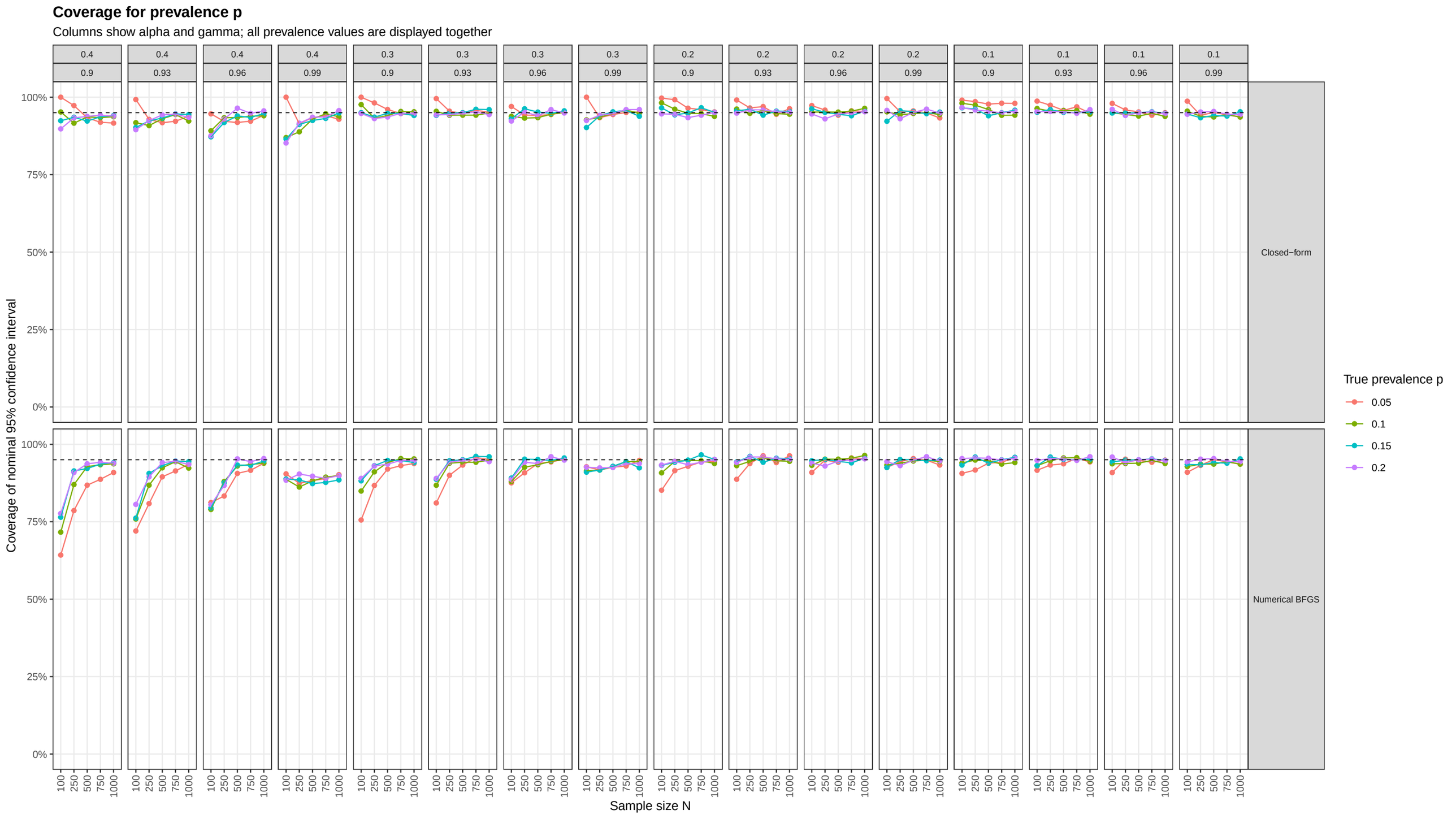
Columns show alpha and gamma; all prevalence values are displayed together

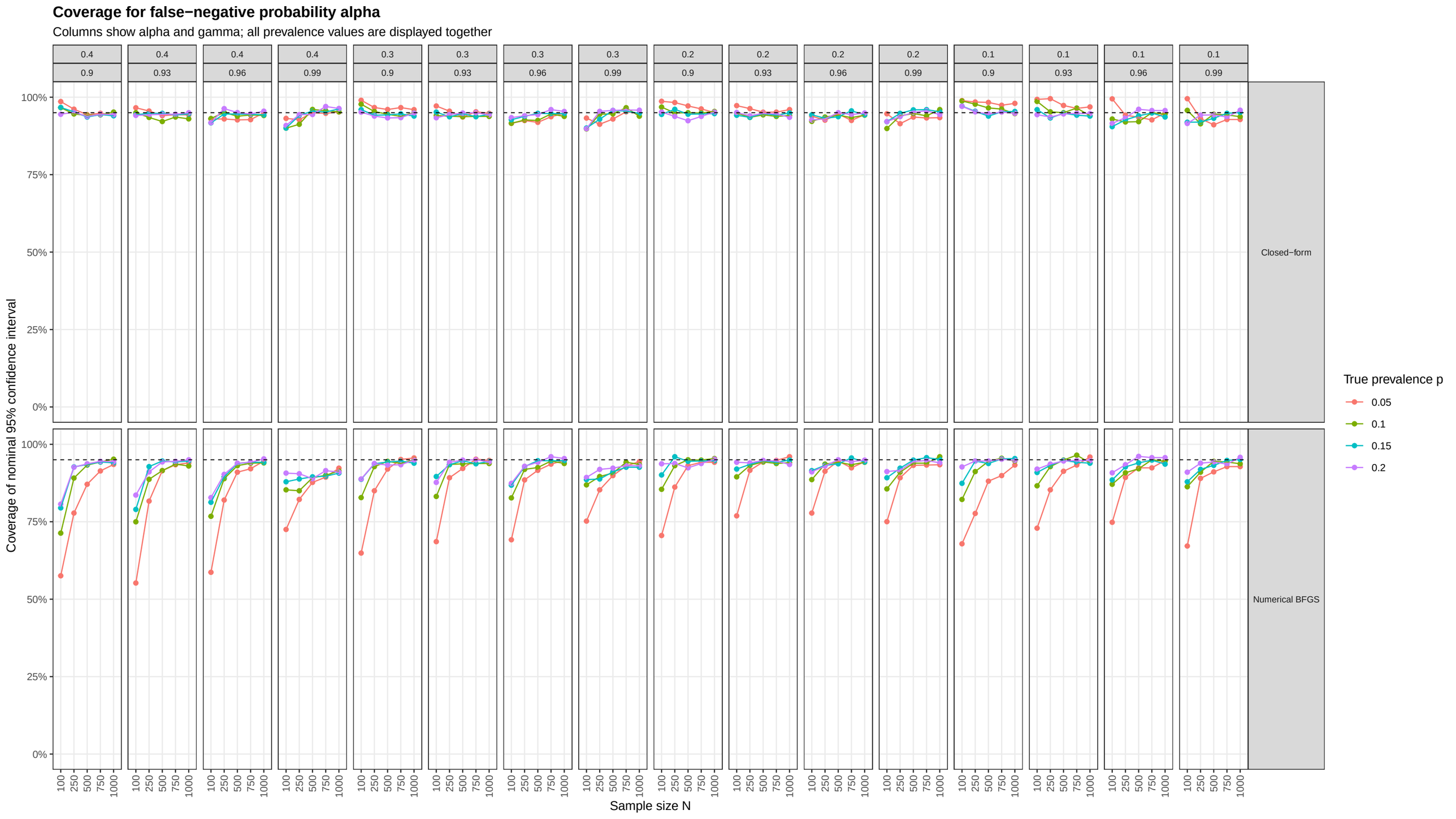


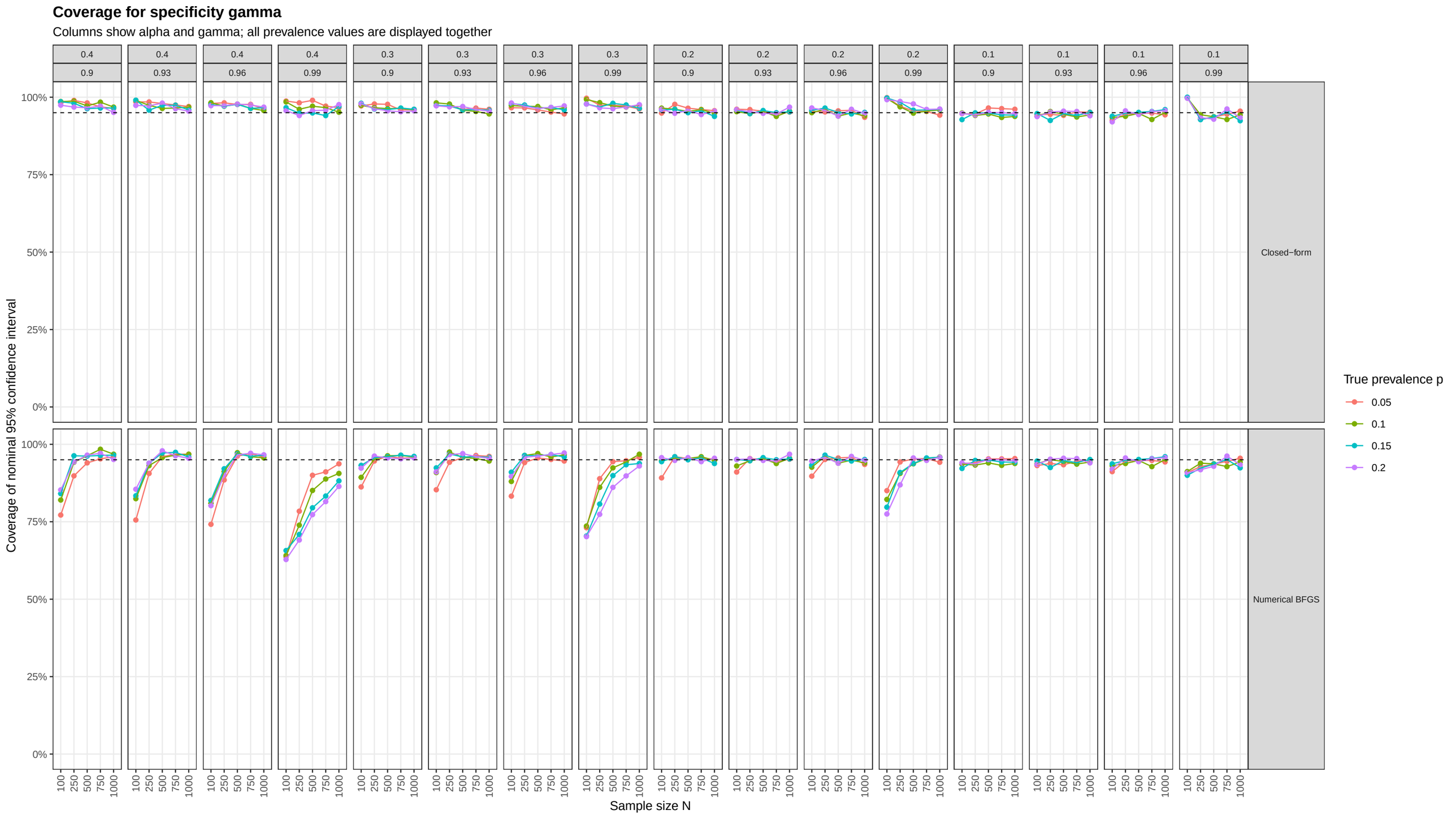
MSE for specificity gamma

Columns show alpha and gamma; all prevalence values are displayed together




Coverage for prevalence p
Columns show alpha and gamma; all prevalence values are displayed together
0.4
0.9
0.4
0.93
0.4
0.96
0.4
0.99
0.3
0.9
0.3
0.93
0.3
0.96
0.3
0.99
0.2
0.9
0.2
0.93
0.2
0.96
0.2
0.99
0.1
0.9
0.1
0.93
0.1
0.96
0.1
0.99
Closed−form
Numerical BFGS
Coverage of nominal 95% confidence interval
100%
75%
50%
25%
0%
100
250
500
750
1000
Sample size N
True prevalence p
0.05
0.1
0.15
0.2


Coverage for false−negative probability alpha
Columns show alpha and gamma; all prevalence values are displayed together
0.4
0.9
0.4
0.93
0.4
0.96
0.4
0.99
0.3
0.9
0.3
0.93
0.3
0.96
0.3
0.99
0.2
0.9
0.2
0.93
0.2
0.96
0.2
0.99
0.1
0.9
0.1
0.93
0.1
0.96
0.1
0.99
Closed−form
Numerical BFGS
Coverage of nominal 95% confidence interval
100%
75%
50%
25%
0%
100
250
500
750
1000
Sample size N
True prevalence p
0.05
0.1
0.15
0.2


Coverage for specificity gamma
Columns show alpha and gamma; all prevalence values are displayed together
0.4
0.9
0.4
0.93
0.4
0.96
0.4
0.99
0.3
0.9
0.3
0.93
0.3
0.96
0.3
0.99
0.2
0.9
0.2
0.93
0.2
0.96
0.2
0.99
0.1
0.9
0.1
0.93
0.1
0.96
0.1
0.99
Closed−form
Numerical BFGS
Coverage of nominal 95% confidence interval
100%
75%
50%
25%
0%
100
250
500
750
1000
Sample size N
True prevalence p
0.05
0.1
0.15
0.2

Inadmissibility rate

Columns show alpha and gamma; all prevalence values are displayed together

